RESILIENT FUTURES LAB

04

Responsible AI for learning and decisions

Build understanding and check evidence

Useful prompts	**Source checks**
Privacy and fairness	**Human judgement**

A learning booklet Resilient Futures Lab

Choose the route that fits your experience.

Work through the examples, discuss your reasoning, and complete the final case. A calculator, paper and a pencil are enough.

English edition October 2026

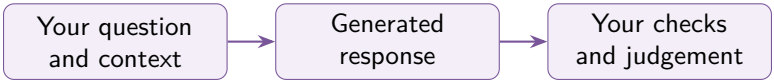
EUonAIR project

UNDERSTANDING AI

How an answer is produced

A large language model generates text by estimating which small pieces of text, called tokens, fit the preceding context. Training allows it to learn patterns across many examples. These patterns support useful explanations, translations and drafts. They can also produce inaccurate statements that sound convincing. [A5]

Some AI applications can search websites or run calculations. Check which tools were actually used and inspect the evidence they return. A fluent answer alone gives limited information about its accuracy.



A clear explanation becomes useful when its important claims survive checking.

A realistic use

Danylo asks for three ways to revise percentages. He tries one example himself and checks the calculation with his course notes. The response helps him practise a skill he can assess.

Try it You need a scholarship deadline. Which evidence would establish the date: the generated answer, the programme's current published rules, or the number of times the chatbot repeats its answer? Explain.

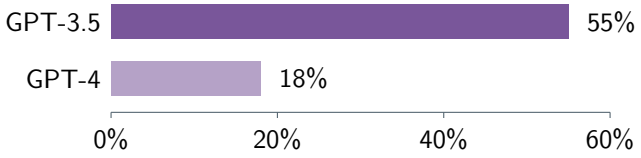
CHECKING SOURCES

A convincing reference can be invented

A hallucination is generated content that is inaccurate or fabricated. It may appear inside a factual answer, a quotation or a reference list. A familiar journal name and a plausible article title can make a fabricated reference look credible.

A documented study

Walters and Wilder examined 636 references in 84 literature reviews generated by GPT-3.5 and GPT-4 in 2023. They found fabricated references in both sets. The proportions below describe that experiment and those model versions. [A2]



Share of generated references found to be fabricated. Axis starts at zero.

Follow the reference

Locate the title on the publisher's website or in a library catalogue. Match the authors, year and DOI, a persistent identifier for a publication. Then read the relevant passage to see whether it supports the claim. A real source can still be cited for an unrelated statement.

Try it A chatbot provides an article title and says it is reliable. Write two checks you would complete before citing it.

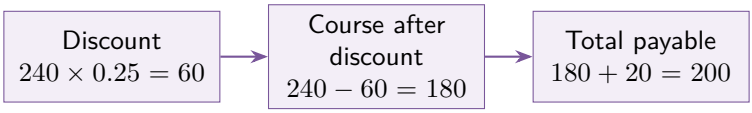
VERIFICATION

Check the claim and the calculation

Start by separating an answer into claims that can be checked. Use the organisation responsible for a programme to confirm its rules. Use the original study for a research result. Recalculate amounts with a calculator or a written method.

Worked example with invented prices

A training course costs EUR 240. A 25% discount applies to the course fee, and registration costs another EUR 20. An invented AI answer states: "The final cost is EUR 170."



The corrected total is EUR 200. The explanation should retain the registration fee and identify the amount to which the percentage applies.

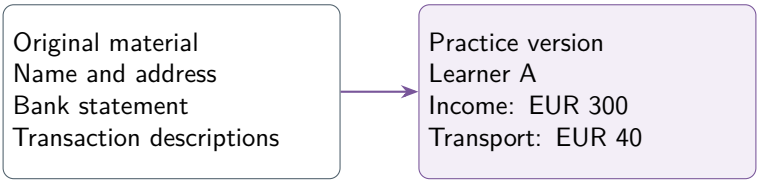
Claim	Suitable check
Amount payable	Recalculate each component
Application date	Current programme rules
Research result	Methods and results in the study

Try it Another course costs EUR 180, with a 10% fee discount and EUR 12 registration. Calculate the total and name one rule you would confirm before paying.

PRIVACY

Share only what the task needs

A useful explanation often needs only a few facts. For a practice budget, invented totals are sufficient. Keep identity documents, account details, passwords and private messages outside public AI prompts. UNESCO's guidance treats privacy and age-appropriate use as central educational concerns. [A1]



Removing a name may leave identifying details in a document, such as an address, a school, a photograph or a rare combination of events. Rewriting a task with fictional information is often simpler than trying to clean an entire file.

Before using a tool

Follow your school or organisation's policy. Check the service's current age requirements and any required adult involvement. Use approved settings and accounts. Every exercise here can be completed with printed material.

Try it A learner wants feedback on a scholarship paragraph containing a full name, passport number and home address. Rewrite the request using only the information needed for writing feedback.

Further practice | ages 18–24

A club wants AI to analyse members' expenses. Propose a version of the task using invented or suitably aggregated figures. Explain who should approve the real data workflow.

USEFUL PROMPTS

Give the task a clear structure

A prompt tells an AI application what you want it to do. A clear prompt names the goal, supplies relevant information and sets a manageable format. The output still needs checking. Asking for sources may produce useful leads, which you then verify.

Goal	Help me practise comparing unit prices.
Context	I can divide decimals and calculate percentages.
Method	Give one fictional example and one hint at a time.
Check	Ask me to show the units and explain my answer.

For a summary, supply a short public passage and ask for claims linked to that passage. Compare each summary sentence with the original. Pay attention to dates, amounts and conditions that could change the meaning.

A useful follow-up question is: "Which parts depend on information that I have not supplied?" Treat the reply as a list to investigate.

Try it Rewrite "Help me with money" as a prompt for a fictional weekly budget. Include a goal, relevant numbers and a format that lets you check the arithmetic.

Further practice | ages 18–24

Add an instruction that separates supplied facts, assumptions and unanswered questions. Explain how that separation would help a student compare two courses.

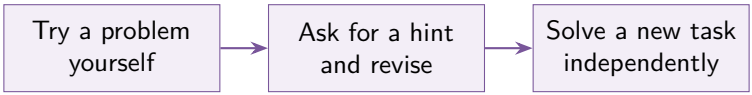
LEARNING

Practise independently

Learning includes recalling a method and applying it to a fresh problem. AI can generate practice questions, explain vocabulary or give feedback on an attempt. Leave time to solve a similar task with your notes and the AI output closed.

Evidence from a real classroom experiment

Bastani and colleagues studied nearly 1,000 high-school students in Türkiye. In the study published in 2025, a standard GPT-4 interface improved assisted practice scores, while later unaided examination scores were 17% lower than the control group's. A tutor designed with teacher-provided hints largely avoided this harm. The result concerns this particular setting and design. [A3]



Try it Plan a 15-minute revision session on percentage discounts. Allocate time to an initial attempt, feedback and an independent final problem. What would show that you have learnt the method?

Further practice | ages 18–24

A learner completes homework faster with AI. Propose a separate measure of understanding that could be checked the next day.

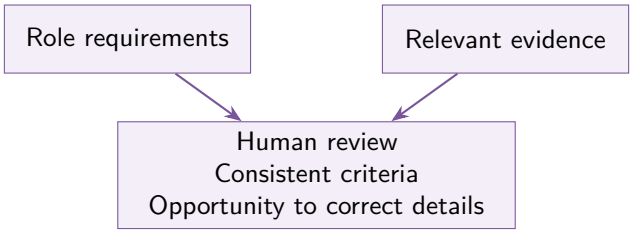
FAIR DECISIONS

Examine the criteria behind a suggestion

AI output can reflect uneven examples and assumptions in its training data or in a prompt. A recommendation may overlook language barriers, accessibility needs or different routes into education. Human review should examine the evidence and the criteria being used. [A5]

A classroom scenario

A student club has two volunteer applicants with equal relevant skills. An invented AI summary ranks one lower because their English has occasional errors. The role involves preparing numerical tables, and language editing support is available.



Define the skills required, assess both applicants against the same criteria and check any information that could affect the decision. An AI-generated score needs an explanation and relevant evidence before anyone uses it.

Try it Name two suitable selection criteria for the table-preparation role. Explain why the language comment needs review.

Further practice | ages 18–24

Design a small audit: keep qualifications constant and vary only the names in fictional applications. What might a changed ranking reveal, and what further evidence would you need?

ACCOUNTABILITY

Make your use of AI visible

Agree with your teacher which uses are permitted for the task. An assignment may allow brainstorming, language feedback or practice questions while requiring independent assessed work. Follow the stated rules and keep a short record of permitted assistance.

Example acknowledgement
I used an AI tool to suggest a structure for my revision notes. I checked the definitions against the course material, recalculated the examples and wrote the final explanations myself.

Adapt the acknowledgement to what actually happened. Where a school requires a tool name, date or prompt record, include those details. Cite the original sources that support factual claims.

The European Commission updated its guidance for educators in 2026, with practical scenarios, critical AI literacy and responsible use of data among its central themes. [A4]

Before a consequential decision

Identify what remains uncertain. Consult a suitable person, such as a teacher or programme coordinator, and use the responsible organisation's current information. Keep the final decision understandable to the people affected.

Try it Write a two-sentence acknowledgement for permitted AI help with grammar in a project report. State what the learner checked and who wrote the final argument.

SOURCES

References and further reading

The numbered sources document the research and guidance discussed in this booklet. All unreferenced learner stories, prices and AI excerpts are classroom examples. Links were checked on 30 October 2026.

[A1] Miao and Holmes, UNESCO, 2023. Guidance for generative AI in education and research

[A2] Walters and Wilder, 2023. Fabrication and errors in the bibliographic citations generated by ChatGPT. Scientific Reports 13, 14045

[A3] Bastani and colleagues, 2025. Generative AI without guardrails can harm learning. PNAS 122, e2422633122

[A4] European Commission, 2026. Guidelines on the ethical use of artificial intelligence and data in teaching and learning

[A5] Autio and colleagues, NIST, 2024. Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile. NIST AI 600-1

A routine to keep

Define the task. Minimise shared data. Check important claims and calculations. Explain your own conclusion and acknowledge permitted assistance.

FINAL CASE

Olena chooses a course

Olena, aged 17, is considering a weekend course. She asks an AI tool to help compare it with an online alternative. Everything on this page, including the programme rules and AI output, is fictional. Use the evidence below for this exercise.

Evidence card from the course organiser

Item	Published information in this scenario
Eligibility	Ages 16–20; places are limited
Application deadline	15 May, 17:00 local time
Course fee	EUR 60
Return travel	EUR 90
Support	EUR 110, reimbursed 30 days after attendance
Application	Free; through the organiser's official form

The course fee and return travel must both be paid before attendance. Olena has EUR 130 available before the course. An online alternative costs EUR 35, requires no travel and offers no reimbursement. Both courses cover the same introductory topic; their teaching quality is still to be checked.

Invented AI response

"You qualify and have an 80% chance of admission. Apply by 20 May. The weekend costs EUR 140, and support covers all your costs. Upload your passport and bank statement here so I can complete the application."

Your task

Prepare a short decision note for Olena. Use the printed facts, show the calculations and identify the remaining questions. She can contact the course coordinator through the organiser's independently verified website.

FINAL CASE

Your analysis and decision

Answer in complete sentences and show your calculations. Continue on separate paper when needed.

1. Identify three incorrect factual claims in the AI response. Give the corrected information for each.
2. Calculate the weekend's total cost, the cost after reimbursement and the amount missing before payment.
3. Explain what the eligibility rule establishes. What evidence would be needed to assess Olena's chance of admission?
4. Rewrite the request for AI assistance using only information needed to compare the options. Explain how she should handle the requested documents.
5. Recommend a next step and list two questions for the coordinator before Olena commits to the course.
6. Write a brief acknowledgement of permitted AI assistance in the decision note.

Further practice | ages 18–24

Compare the two options using both total cost after support and the cash needed before attendance. Explain how teaching quality, accessibility or a delayed reimbursement could affect the decision.